

Assignment 2 (AI)

Deadline: 14th April 2025

Q1. Design a Sudoku puzzle where the board consists of 81 squares, some of which are initially filled with digits from 1 to 9. The puzzle is to fill in all the remaining squares such that no digit appears twice in any row, column, or 3×3 box. A row, column, or box is called a unit.

- a. Represent the Sudoku problem in CSP by identifying the variable, domain, and constraint.
- b. Implement the problem using backtracking search, what is avg. time taken by the algorithm for 10 runs.
- c. Analyse how different fault finding algorithms such as Forward Checking, Arc consistency improve the computational time of backtracking search?
- d. Analyse how different Heuristics MRV (Minimum Remaining Values), Degree heuristic, Least Constraining Value affect the computational time of backtracking search?

Note: Prepare Comparison table in your report and provide the reasons for performance improvements.

Q2. Consider a simple N-Queens problem. Take a suitable value of N and write a Python code for simulated annealing to solve the problem. As we know, if we take $T=0$ in SA, it falls back to simple hill climbing search. Modify the SA algorithm in order to make it an HC solution. In the report, write proper theoretical justification in support of your modification (e.g., calculation of the acceptance probability) and the behavioral difference between the two approaches. (Note: No marks will be given for simply implementing the HC solution.)